



EU Cloud Sovereignty Framework

VMware Cloud Foundation 9.1

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Disclaimer

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Version

910-20260623-01

Introduction

This document describes how VMware Cloud Foundation (VCF) 9.1 and the separately licensed advanced services assessed alongside it, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, and VMware Avi Load Balancer, can support organizations addressing the sovereignty objectives of the European Commission's Cloud Sovereignty Framework. The framework, published by the Directorate-General for Digital Services (version 1.2.1, October 2025), defines eight Sovereignty Objectives covering strategic, legal and jurisdictional, data and AI, operational, supply chain, technology, security and compliance, and environmental sustainability dimensions, assessed through Sovereignty Effectiveness Assurance Levels (SEAL-0 through SEAL-4) in EU cloud procurement. The framework assesses cloud service providers; this guide describes how an organization operating VMware Cloud Foundation, such as a European cloud provider or a public-sector private cloud, can draw on platform capabilities when addressing each objective. The framework has no SCF crosswalk; the objective-to-SCF mapping in this guide was authored for the VCF assessment. Each assessed product is reported separately for every control, with VCF covering the full platform, including its consumption services and Private AI Services capabilities. The responsibility for demonstrating sovereignty assurance levels in a procurement remains with the tendering cloud service provider.

https://commission.europa.eu/document/download/09579818-64a6-4dd5-9577-446ab6219113_en

This guidance evolves. Please check the current revision at: <https://brcm.tech/vcf-compliance>

Additional Resources

This is one of a library of companion guides that map VMware Cloud Foundation and its advanced services to security and compliance frameworks. For each framework, the guide is published as a spreadsheet, a CSV data file, a Markdown document, and a PDF, together with a Security Configuration Guide crosswalk that identifies the VCF technical hardening settings relevant to that framework. The companion guide library, the VMware Cloud Foundation Security Configuration Guide, and related security documentation, sample scripts, and Q&A are available at <https://brcm.tech/vcf-security>.

Third Party Identifiers and Mappings

This document includes regulatory compliance and security control identifiers from external sources as a convenience to end users. This does not constitute endorsement, in either direction.

There is not a one-to-one mapping of product capabilities to third-party controls. A product capability, or set of capabilities, may be applicable to multiple controls. Conversely, a control may be satisfied with the use of multiple capabilities.

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<https://securecontrolsframework.com/>

Coverage Levels

Each control entry below carries the framework control text, the SCF control or controls it maps to, and an Aggregate Coverage line showing the strongest position any one of the assessed products holds for that control. Products that meet or contribute to the control follow as subsections, each with its own coverage level and response. Products the control does not apply to are listed together at the end of the entry, with a brief explanation. The coverage levels are:

- **Meets** -- the product directly provides the capability described in the control. No additional organizational process, external product, or human activity is needed beyond configuring the product itself.

- **Contributes** -- the product provides technical capabilities that help satisfy the control, but the control also requires something outside the product to be fully met, such as an organizational process, human activity, or an additional tool.
- **Not Applicable** -- the control does not apply to the product. Physical facility controls, staffing decisions, organizational governance processes, and domains the product does not operate in fall here.
- **Not Supported** -- the control is relevant to the product's domain (infrastructure, networking, security), but the product lacks the specific capability the control requires.
- **Not Assessed** -- the control has not yet been evaluated against the product.

Controls

Control SOV-1

Strategic Sovereignty: Strategic sovereignty captures the degree to which the services of a cloud provider (or technology actor) are anchored within the European Union legal, financial, and industrial ecosystem. It assesses ownership stability, governance influence, and alignment with EU strategic priorities. Contributing factors: Ensuring that bodies having decisive authority over your services are located within EU jurisdiction; Evaluating the assurances against change of control; Degree to which the provider relies on financing coming from EU sources; Extent of investment, jobs, and value creation within EU; Involvement in EU initiatives, Consistency with digital, green, and industrial sovereignty objectives defined at EU level; Ability to sustain secure operations against requests to cease or suspend the service, or if vendor support is withdrawn or disrupted.

SCF Controls: BCD-01, BCD-14, GOV-12, TDA-20.3, TPM-04

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control SOV-2

Legal & Jurisdictional Sovereignty: Legal & Jurisdictional sovereignty evaluates the legal environment, exposure to foreign authority, and enforceability of rights that govern the services of a technology provider. It determines the extent to which the services are anchored in European jurisdiction and insulated from external legal claims. Contributing factors: The national legal system governing the provider's operations and contracts; Degree of exposure to non-EU laws with cross-border reach (e.g., US CLOUD Act, Chinese Cybersecurity Law); Existence of legal, contractual, or technical channels through which non-EU authorities could compel access to data or systems; Applicability of international regimes, which may restrict usage or transfer; Location of intellectual property creation, registration, and development (EU vs. third countries), legal jurisdiction where IP is created and developed.

SCF Controls: CLD-09, CPL-01, DCH-14.2, DCH-26, TPM-04.4

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides several layers of technical capability that help organizations manage data placement and enforce geographic constraints, which is relevant when addressing digital sovereignty requirements that mandate data localization within specific jurisdictions.

VCF Automation organizes infrastructure into regions, and a region quota defines the compute and storage quota that an organization receives against a given region. The provider administrator decides from which regions to allocate quota to each organization, giving centralized control over where workloads and their associated data can be provisioned. Image mappings are also region-aware, with available images determined by the selected region. This region-based architecture allows administrators to constrain resource provisioning to specific geographic locations, supporting data residency requirements. Cloud template location constraints can be applied to both Machine and Disk resources to separate disk placement across locations, giving fine-grained control over where compute and storage resources are placed within a deployment.

Constraint tags in cloud templates further refine placement behavior. Tags added to projects and cloud templates function as constraint tags when matched against capability tags on infrastructure resources, profiles, and cloud zones. Project-level constraint tags take precedence over cloud template constraint tags at deployment time, allowing cloud administrators to enforce governance rules centrally. For example, if an administrator creates a location:london tag on a project but a developer places a location:boston tag on a cloud template, the project tag takes precedence and the resource is deployed to infrastructure containing the location:london tag. Constraint tags can also use variables to populate tags at request time, enabling flexible location selection within administrator-defined boundaries.

vSphere VM Storage Policies provide an additional placement control mechanism at the virtual machine level. Tag-based rules within VM Storage Policies allow administrators to filter datastores by geographic location, including or excluding datastores based on location-encoded tags. Datastores not represented by storage providers can be tagged within VM Storage Policies to communicate geographic properties that are not available through the storage provider interface, allowing geographic placement constraints to be enforced at the storage policy layer.

Policy as Code in VCF Automation enables organization administrators to enforce instance-centric policies for Infrastructure as a Service resources from a central location. These policies can restrict where workloads are deployed and what resources they can consume, providing a programmatic mechanism for enforcing placement constraints that align with data localization obligations.

For data protection at rest, vSAN provides cluster-based data-at-rest encryption. vSAN encryption requires a Key Management Server (KMS) and supports integration with a KMS cluster for encryption key management. vSphere Native Key Provider is also available as an alternative for encrypting virtual machines in vSAN clusters. The separation of encrypted data from key material allows organizations to place encrypted workloads in a required jurisdiction while maintaining control of

encryption keys in a separate, trusted location, limiting the exposure of data at rest to enforcement actions in a single jurisdiction.

For multi-site deployments, NSX Federation provides a topology where a Global Manager coordinates with Local Managers at each site. On a Local Manager, administrators can see local objects and any global objects that apply to that location, while the Location Manager monitors communication details between Local Manager and Global Manager. Security objects in NSX Federation have a region that can be Location (automatically created for each location with that location's span), Global (spanning all available locations), or Custom Region (spanning a subset of available locations). This architecture gives operators location-aware visibility and control over both networking and security configuration at each site, which is relevant when different jurisdictions impose different requirements on network traffic and data flows.

VCF Operations provides location-awareness features that help track where infrastructure resides. A GEO Location tag type allows tag values to be identified with a location on a world map, and administrators can select a physical location on a map for each data center with search and zoom capabilities. Objects assigned a GEO Location tag appear on a geographical map in VCF Operations, and the Geo Widget shows the health of each object at each location. These tagging and mapping features support inventory and compliance tracking for geographically distributed infrastructure.

Organizations should also be aware that cross-zonal datastores typically replicate data across zones. When data sovereignty requirements prohibit cross-border replication, storage architecture decisions must account for this behavior to avoid unintended data movement outside permitted jurisdictions.

Defining organizational data residency policies, mapping regulatory requirements to specific jurisdictions, managing contractual obligations around data localization, and determining which jurisdictions apply to which workloads are organizational responsibilities outside VCF's scope.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to constraining the impact of digital sovereignty laws through GeoIP visibility, IDS/IPS-based geographic detection, distributed firewall scoping, and federated policy management at the network layer.

vDefend's Security Services Platform includes GeoIP Monitoring, which displays a map of network traffic to and from every country in the world. This capability requires the Security Services Platform to be deployed with the metrics feature enabled and gives administrators visibility into geographic traffic patterns across the environment. vDefend IDS/IPS signature rules support geoip matching on source IP addresses, destination IP addresses, or both, providing a detection mechanism that identifies traffic associated with specific countries within east-west distributed traffic flows. This allows administrators to detect and respond to data flows that cross jurisdictional boundaries where policy requires them to remain local.

vDefend supports policy scoping across global, regional, and local spans in a federated deployment. Security Services Platform regions can be associated with Security and Network policies to control policy scope across multiple locations. DFW policies applied globally allow sources or destinations to be any object without span restrictions, while the Applied To field scopes enforcement to specific zones, tenants, or applications without affecting policies defined for other workloads. This regional policy architecture allows organizations to define distinct security controls for workloads in different jurisdictions while maintaining a consistent baseline across sites. In a federated environment, malicious traffic can be blocked across Local Managers from the Global Manager, providing centralized enforcement across geographically distributed sites.

These network-layer capabilities complement the broader VCF data sovereignty mechanisms, including encryption and workload placement constraints, by providing traffic-level controls and visibility that support geographic boundary enforcement. Defining organizational data residency policies, mapping regulatory requirements to specific network policies, and managing contractual obligations around data localization remain organizational responsibilities outside vDefend's scope.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) supports data localization through customer-controlled site pairing and explicit replication targeting. PNR operates within customer-owned and customer-managed infrastructure, replicating only between explicitly paired VMware vCenter sites that an administrator configures. Protection groups define which virtual machines replicate to which recovery site, and replication between sites requires deliberate administrator action.

PNR supports deployment across multiple geographic regions, with PNR Server and vCenter instances at each site running in locale-specific configurations. Documented topologies include a Japanese locale site paired with a United States English locale site, with each PNR Server extending the vCenter Appliance instance at its own site. This allows organizations to align recovery infrastructure with jurisdictional boundaries and keep replicated workloads within specified regions.

Site pairing is configured between two specific vCenter instances, each running on infrastructure the customer physically controls. Inventory mappings between sites are configured manually, mapping networks, folders, compute resources, and datastores from the protected site to corresponding objects on the recovery site. An optional reverse mapping capability creates the equivalent mappings from the remote site back to the local site to support re-protection. Manual storage policy mapping allows multiple storage policies on the local site to be mapped to a single storage policy on the remote site. These administrator-driven mappings keep placement decisions for protected and recovered workloads within the customer's control.

For PNR deployments that integrate with VCF Automation, a content library must be configured and available for upload in each region where PNR is being deployed, reinforcing the per-region scoping of recovery infrastructure. The VCF Operations Orchestrator plug-in for PNR provides workflows for listing and managing network, folder, and resource mappings on a per-site basis, supporting programmatic verification that data placement aligns with jurisdictional requirements.

Cyber Recovery, the ransomware recovery capability included in PNR, uses data seeds, which are pre-populated datasets that jumpstart replication in a new location by providing the target with an initial copy of source data. Protection groups allow administrators to selectively replicate workloads, so data subject to localization requirements can be excluded from cross-border replication or directed only to recovery sites within the same jurisdiction.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) contributes to this control through role-based access controls, identity integration, and connection-level authorization mechanisms that restrict data transfer operations to authenticated and authorized entities.

DSM provides two platform-level roles: DSM Admin and DSM User. These roles govern what operations a user may perform, including provisioning databases and establishing replication relationships between instances. Users can be sourced from LDAP or LDAPS identity providers, or maintained as local accounts. When directory service integration is configured, DSM uses the configured directory service for both authentication and authorization. Administrators import LDAP users through the DSM Admin UI and assign them the DSM Admin or DSM User role. MySQL database instances can further delegate authentication to Active Directory via LDAPS. SQL Server databases support Windows Principal authentication, allowing users to authenticate with Active Directory credentials; administrators can optionally configure privileged Active Directory account credentials to activate automated Kerberos integration. Role assignment determines which users are permitted to initiate operations that involve transferring data, such as establishing replication or running migrations.

For client applications and services connecting to DSM-managed database clusters, DSM enforces certificate-based trust requirements before connections can be established. Custom trusted certificates used in DSM must comply with FIPS requirements. Database clusters issue certificates for client connectivity and monitoring connections. PostgreSQL databases support host-based authentication (HBA) configuration, with the default configuration requiring SCRAM-SHA-256 authentication for service accounts including `postgres_exporter` and `pgadmin`. When LDAPS users access PostgreSQL databases, administrators configure specific privilege grants, including database and schema-level access, before those users can perform operations. These mechanisms control which client systems and identities may establish connections and exchange data with the managed databases.

Data transfer operations between DSM database instances carry additional authorization requirements tied to role and identity verification. During PostgreSQL database migration, DSM requires TLS verification between source and target clusters. For multi-site PostgreSQL replication, users with the DSM User role must manually establish trust between primary and secondary instances before replication proceeds; users with the DSM Admin role have trust established automatically. This role distinction governs which identities may initiate and complete cross-instance data transfers. MySQL database migration to DSM requires creation of a dedicated user account with minimum privileges of `BACKUP_ADMIN`, `REPLICATION_SLAVE`,

SELECT, and INSERT on the mysql schema, restricting the migration account to only those permissions the operation requires. DSM also verifies network access to the source database and validates SSL trust during MySQL replication operations.

For data forwarded to external systems, DSM requires authenticated connections before transfer proceeds. Metrics forwarding supports Mutual TLS (mTLS) authentication, configured by enabling the Client Certificate Authentication option and supplying the client certificate and private key. Forwarding to other metrics targets requires verification of server address, port, authentication type, credentials, and certificates before the forwarding connection is accepted. Webhook notification endpoints support basic authentication with Authorization User and Password credentials, bearer token authentication via Authorization header, or authorization embedded in the webhook URL for Slack-compatible endpoints. For self-service database consumption, the Consumption Operator requires a TLS certificate for secure communication with the DSM provider, and an authentication secret containing the credentials needed to connect to the DSM provider must be created and configured before the operator can connect.

DSM's authorization mechanisms primarily govern user-initiated operations and connection-level access within the database service layer. Verifying authorizations for all Technology Assets, Applications and/or Services transferring data across interconnected systems requires organizational data governance processes that extend beyond DSM's scope. Organizations must define which services are authorized to connect to DSM-managed databases, maintain records of authorized interconnections, and audit application-layer data flows independently of the platform controls DSM provides.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides several application-layer mechanisms that support data localization requirements by controlling where application traffic is directed and which clients can access specific services.

The Global Server Load Balancing (GSLB) feature in Avi includes a geolocation-based load balancing algorithm that directs client requests to region-specific application deployments based on the geographic origin of the request. Avi maintains a geolocation database on its Controllers, which can be loaded from file objects of type GSLB Geo DB and used as a Geo profile for the geolocation-based algorithm. The GSLB geolocation database supports multiple input files with a priority mechanism that determines which source takes precedence when multiple databases provide data for the same IP address. Geolocation database records include IP address range start and end, latitude, longitude, and an optional name field representing city, region, and country in a hierarchical format.

Avi's DNS Policy feature provides granular DNS-level traffic steering and filtering based on geographic criteria. DNS policies can match on client IP addresses, protocol, query name, query type, or geographic location, and apply actions including allowing or blocking DNS queries, rate limiting, or selecting a GSLB site. DNS Policy rules support geolocation-based matching criteria and GSLB group selection actions, enabling country- and region-level enforcement directly in the DNS resolution path. DNS Policy can block DNS queries from a particular geographic area by configuring a client IP match rule with IP addresses associated with that area and setting the rule action to drop.

For custom programmable logic, Avi DataScript provides the `avi.utils.get_geo_from_ip` function, which retrieves geolocation information including autonomous system number, country code, country name, continent name, continent code, and geographic coordinates (latitude, longitude, region, and city) for a given IP address. Avi DNS DataScript also supports geo-based DNS response routing, allowing different A record responses to be returned based on the client's country location. This enables organizations to implement dynamic, policy-driven restrictions on which clients can reach specific application endpoints.

Avi's Web Application Firewall (WAF) integrates with the geolocation database. The WAF Policy can be configured with a Geo Location Mapping Database, and the Avi Service Engine supports geolocation database usage in HTTP and network policies at region and city levels, not limited to country-level granularity. Geo-mapping attributes including `ATTRIBUTE_COUNTRY_CODE` and `ATTRIBUTE_ISP_NAME` can be referenced in security policies to apply geo-based access controls. The Geo-DB mapping feature can group geographic regions such as APAC countries or embargoed countries for use in policy matching and enforcement across multiple rules.

These capabilities operate at the application delivery layer. Data localization governance also requires organizational policies governing data classification, workload placement decisions at the VCF platform level, and procedures for responding to enforcement actions under applicable laws.

Control SOV-3

Data & AI Sovereignty: Data & AI sovereignty focuses on the protection, control, and independence of data assets and AI services within the EU. It addresses how data is secured, where it is processed, and the degree of autonomy customers retain over AI capabilities. Contributing factors: Ensuring that only the customer, not the provider, has effective control over cryptographic access to their data; Visibility into when, where, and by whom data is accessed, including auditability of AI model usage, mechanisms guaranteeing irreversible removal of data, with verifiable evidence; Strict confinement of storage and processing to European jurisdictions, with no fallback to third countries; Extent to which AI models and data pipelines are developed, trained, hosted, and governed under EU control, minimizing dependency on non-EU technology stacks.

SCF Controls: AAT-10.17, AAT-10.19, CLD-09, CRY-01, CRY-09, CRY-09.4, CRY-09.7, DCH-09, DCH-09.1, DCH-26, IAC-21.4, MON-02.7, MON-03.2, RSK-09.2

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides provenance tracking mechanisms for model artifacts and data sources used by AI workloads. Models are stored in Harbor registries integrated with PAIS as a model storage backend, with authentication and certificate-based trust required for model retrieval. PAIS is also compatible with OCI-compliant registries from other vendors, allowing organizations to use existing model storage infrastructure while preserving artifact lineage. Each model artifact stored as an OCI object carries an immutable manifest and a unique content digest, which provides verifiable lineage from a deployed model back to its specific revision and the registry of origin.

In disconnected (air-gapped) environments, a Harbor registry acts as a controlled intermediary between external model sources and the internal deployment environment. The artifact mirroring tool (AMT), distributed as part of the pais CLI plug-in, helps administrators populate the Harbor registry from external sources before catalog publication. PAIS requires HTTPS trust bundles configured for the OIDC provider, the Harbor registry, content management systems hosting knowledge base data, and Model Context Protocol servers providing tools to models, which establishes verified channels for the artifact and data sources entering the platform.

For knowledge base data used in retrieval-augmented generation, PAIS Data Indexing and Retrieval connects to enterprise content management systems such as Microsoft SharePoint sites and Amazon S3 compatible stores as data sources. Data sources are created with a specific type that cannot be changed after creation, and a knowledge base can be linked to a maximum of ten data sources, with one data source linkable to multiple knowledge bases. PAIS tracks document-level changes during indexing operations, capturing counts of crawled documents, new documents found, deleted documents, existing unmodified documents, modified documents requiring embeddings update, and skipped documents. Indexing operations also record megabytes downloaded, number of new chunks indexed, final index size after removal of deleted documents, and timing metrics for download, embedding generation, and the index operation. PAIS supports OpenTelemetry collector integration for metrics collection and large language model traces, including knowledge base indexing traces, which records observable evidence of content provenance at the data layer.

PAIS gives a foundation for content provenance through OCI-based artifact immutability and digest-based integrity verification, combined with knowledge base indexing telemetry. The platform does not implement a formal content provenance benchmarking standard such as SLISA (Supply Chain Levels for Software Artifacts) or SPDX (Software Package Data Exchange). Organizations requiring benchmarking against industry-recognized provenance standards layer those frameworks on top of PAIS-native artifact and data source management.

VMware vDefend (Contributes)

VMware vDefend (vDefend) contributes to constraining the impact of digital sovereignty laws through GeoIP visibility, IDS/IPS-based geographic detection, distributed firewall scoping, and federated policy management at the network layer.

vDefend's Security Services Platform includes GeoIP Monitoring, which displays a map of network traffic to and from every country in the world. This capability requires the Security Services Platform to be deployed with the metrics feature enabled and gives administrators visibility into geographic traffic patterns across the environment. vDefend IDS/IPS signature rules support geoip matching on source IP addresses, destination IP addresses, or both, providing a detection mechanism that identifies traffic associated with specific countries within east-west distributed traffic flows. This allows administrators to detect and respond to data flows that cross jurisdictional boundaries where policy requires them to remain local.

vDefend supports policy scoping across global, regional, and local spans in a federated deployment. Security Services Platform regions can be associated with Security and Network policies to control policy scope across multiple locations. DFW policies applied globally allow sources or destinations to be any object without span restrictions, while the Applied To field scopes enforcement to specific zones, tenants, or applications without affecting policies defined for other workloads. This regional policy architecture allows organizations to define distinct security controls for workloads in different jurisdictions while maintaining a consistent baseline across sites. In a federated environment, malicious traffic can be blocked across Local Managers from the Global Manager, providing centralized enforcement across geographically distributed sites.

These network-layer capabilities complement the broader VCF data sovereignty mechanisms, including encryption and workload placement constraints, by providing traffic-level controls and visibility that support geographic boundary enforcement. Defining organizational data residency policies, mapping regulatory requirements to specific network policies, and managing contractual obligations around data localization remain organizational responsibilities outside vDefend's scope.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) and vSphere Replication maintain control of cryptographic keys for encrypted material stored or transmitted across sites through the VCF platform's layered key management architecture.

PNR and vSphere Replication require matching key provider configurations at both protected and recovery sites. Encrypted virtual machines cannot be replicated to a remote site unless that site has access to the appropriate encryption keys through a common KMS or vSphere Native Key Provider. This requirement means encrypted material cannot be moved to an external system without the organization explicitly providing key access, maintaining organizational control over the key-to-data relationship. When using the vSphere Native Key Provider, the provider ID configured on the local site must match the provider ID on the remote site, confirming key identity before replication proceeds.

When replication seeds are used, replica disks can have different encryption keys from the source disks, but the key management infrastructure at the recovery site still controls access to those keys. The vSphere Replication appliance automatically installs an encryption agent on source ESX hosts for replication encryption, with key material managed through the platform's key infrastructure rather than embedded in the replication stream. When replication of an encrypted virtual machine is configured, encryption of transferred data is automatically enabled and cannot be disabled, so encrypted machines do not transit unprotected across site boundaries.

The VCF platform provides the underlying key separation architecture. Key Encryption Keys (KEK) wrap Data Encryption Keys (DEK), and the KEK is not stored on the ESX host, maintaining separation between encryption keys and encrypted data. Internal keys are kept in memory only during operations. Fine-grained Cryptographic Operations privileges control which identities can perform key operations, and the vSphere Native Key Provider operates without sending keys to an external system. Site Recovery Manager (SRM) supports FIPS mode activation, which restricts cryptographic operations to approved algorithms through the BouncyCastle FIPS provider.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) provides audit logging at both the platform and database levels, giving administrators visibility into privileged actions taken across the system.

At the platform level, DSM generates system audit events for all user actions performed through the DSM UI and API, as well as actions that DSM itself initiates. These events are accessible through the System Audit Events view in the Provider interface, which is the only DSM UI location for tracking non-database-related operations and tasks. Each audit log entry records an Event Type field that identifies the category of operation performed, an Event Details column that captures operation-specific

information, and a Source field that distinguishes user-initiated events (recorded by the user's email address) from events initiated by DSM without user intervention (recorded as "System"). Audit event data can be exported in CSV and XLSX formats for offline review or integration with external audit processes. Gateway Audit Logs provide an additional layer of visibility into gateway-level operations, with export capability covering all events visible to the reviewing user regardless of pagination limits.

At the database level, PostgreSQL instances provisioned by DSM include the pgaudit extension, which delivers audit logging functionality across all supported PostgreSQL versions. This extension supports auditing of database-level privileged operations including stored procedures and other SQL-level actions. For MySQL databases, DSM enables the `log_bin_trust_function_creators` configuration setting, which governs logging behavior for stored function creators and their associated binary log entries. For SQL Server databases, DSM supports the standard T-SQL `CREATE AUDIT` and `CREATE AUDIT SPECIFICATION` DDL statements for defining custom database audits. DSM can also automatically create a simple audit for login events based on the Login Auditing option selected when creating a SQL Server cluster. The cluster administrator login is granted restricted read-only privileges and specific `ALTER` privileges for server management and auditing, limiting the scope of administrative access within SQL Server.

DSM Administrator and User roles can generate database VM logs for monitoring and audit purposes. The Permissions view in the Administrator UI supports the creation, monitoring, and management of configured users, helping administrators identify which accounts hold elevated access within the DSM environment.

DSM provides the technical mechanisms to capture audit records of privileged actions at the platform and database levels. Organizations remain responsible for defining what constitutes a privileged function for their specific environment, establishing processes for reviewing audit records, configuring forwarding to a centralized log management or SIEM system, and assigning responsibility for ongoing audit review. The audit trail DSM produces supports those organizational processes but does not substitute for them.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) provides mechanisms that help organizations maintain control of cryptographic keys for TLS certificates and encrypted material, through Hardware Security Module (HSM) integration, external key management service support, and role-based access controls over private key handling.

Avi Controller integrates natively with Thales Luna Network Hardware Security Modules for cryptographic key storage and management. When HSM integration is configured, the Controller acts as a client of the HSM and can create RSA and EC key pairs and certificates directly on the HSM device, keeping private key material in hardware under the organization's control rather than in the Avi software configuration store. High availability across multiple HSM devices is optionally configurable. The HSM settings are accessible through the Controller's security certificate management interface.

For cloud deployments, Avi supports customer-managed keys through external key management providers. In AWS environments, the Controller supports customer-managed Customer Master Keys (CMKs), which the organization creates, manages, and rotates. KMS key access can be scoped to a specific key ID or governed through the KMS key policy directly, restricting permissions to the fewest principals necessary. An IAM policy template is available to create and attach the required KMS access policy to the appropriate IAM role. For SNS-SQS encryption, the key policy must grant the SNS service permission to use the CMK with the required KMS actions.

In GCP environments, Avi accepts customer-supplied Cloud KMS key IDs for encryption of Service Engine disks, Service Engine images, GCS buckets, and GCS objects, each configurable with a separate key in the GCP Cloud Configuration section of the deployment settings. This allows the organization to maintain distinct KMS keys for each storage resource type under its own key management infrastructure.

For TLS private keys stored in the Avi Controller configuration store, Avi's RBAC system controls which administrators have the ability to export private keys, restricting that capability to the fewest administrators necessary. This is managed through the Controller's user and role administration interface. Avi also supports importing encrypted private keys via CLI for certificate installation.

Avi Controller allows selection of custom certificates for the encrypted management channel between the Controller and Service Engines, giving organizations control over the certificates used for internal control-plane communication. For VCF deployments, Avi Controller SSL/TLS certificates support both elliptic curve cryptography (EC) and RSA algorithms, with EC recommended.

For Kubernetes environments, the Avi Kubernetes Operator (AKO) supports configuration of both RSA and ECC signed certificates simultaneously on a virtual service through the `alt.crt` and `alt.key` fields, providing flexibility in certificate algorithm selection for ingress workloads.

Avi Controller and Service Engine support configuration of SSH key exchange algorithms, allowing administrators to exclude weaker key exchange methods for management connections.

When deployed in FIPS mode, Avi uses a FIPS 140-3 validated cryptographic module, which restricts cryptographic operations to approved algorithms and requires stricter handling of keys, authentication data, and random number generation.

Control SOV-4

Operational Sovereignty: Operational sovereignty measures the practical ability of EU actors to run, support, and evolve a technology independently of foreign control. It focuses on continuity of operations, skill availability, and resilience against external dependencies. Contributing factors: Ease of migrating workloads or integrating with alternative EU- controlled solutions without vendor lock-in; Capacity for EU operators to manage, maintain, and support the technology without requiring non-EU vendor involvement; Existence of an EU-based talent pool with the expertise to operate and sustain the service; Assurance that operational support is delivered from within the EU and subject exclusively to EU legal frameworks; Availability of full technical documentation, source code, and operational know-how enabling long-term autonomy; Location and legal control of critical suppliers or subcontractors involved in service delivery.

SCF Controls: AAT-20.2, BCD-01, CLD-07, CLD-08, OPS-01.1, PRI-06.6, TDA-20.3

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides several technical capabilities that support the use of industry-recognized virtualization formats, standardized deployment mechanisms, and change-tracking processes, helping organizations maintain interoperability across their cloud infrastructure.

VCF natively supports the Open Virtualization Format (OVF), an industry standard that describes virtual machine image metadata in an XML format. OVF provides a vendor-neutral, portable way to package and distribute virtual machines, enabling interoperability between different virtualization platforms and cloud environments. OVF templates can be captured from virtual machines or vApps and stored in content libraries for consistent deployment. Content libraries support importing .ovf and .ova files, and the deployment specification for these templates supports optional guest operating system and hardware customization specifications applied during the deployment process. OVF templates from content libraries also become available in Kubernetes namespaces as VM images, allowing DevOps teams to self-service virtual machine provisioning through standard Kubernetes tooling. VMware Kubernetes Service (VKS) distributes Kubernetes node images as vSphere Kubernetes releases (VKrs) in the Open Virtualization Appliance (OVA) format, versioned according to standard Kubernetes versioning conventions. Each VKr documents its OS base (PhotonOS or Ubuntu) together with any vSphere-specific customizations and optimizations, making the scope of custom changes explicit and reviewable; legacy VKrs carry explicit backward-compatibility bounds to prior vSphere versions rather than leaving compatibility scope implicit. The VCF Image Baker supports customization of virtual machine hardware version through configurable OVF properties at build time, allowing organizations to produce node images aligned with their hardware baseline requirements. Guest operating system customization specifications are also stored in XML format and can be exported and imported through the VMware vCenter Customization Specification Manager, providing a portable mechanism for sharing standardized guest OS configurations across vCenter instances.

VCF enforces standardized virtual hardware through a defined set of hardware versions. The `HardwareTypes.Version` class defines the valid hardware versions available for virtual machine configuration, and virtual machine CPU and memory configuration ranges are constrained based on the guest operating system and the selected virtual hardware version. Administrators can set specific hardware versions on virtual machines to standardize testing and deployment across the environment, providing a consistent baseline for workload portability.

For container orchestration, VCF provides VMware Kubernetes Service, which uses the standard Kubernetes API and supports standard Container Network Interface (CNI) plugins. NSX integrates with Kubernetes clusters through the Antrea CNI and NSX Container Plugin (NCP), with NCP running in a container and communicating with NSX Manager and the Supervisor control plane. VCF supports multiple container cluster types including Kubernetes, OpenShift, AKS, EKS, GKE, and TKGM, providing broad interoperability with external container platforms. VKS ClusterClass resources carry

kubernetes.vmware.com/max-version-supported and kubernetes.vmware.com/min-version-supported labels, making version compatibility constraints explicit and inspectable through standard Kubernetes tooling; administrators can list available VKS releases and review these constraints using the command-line interface before performing cluster updates. The Kubernetes cloud-controller-manager pattern, which VKS follows, isolates provider-specific code from core Kubernetes components through the CloudProvider interface, creating an architectural boundary that separates VMware-specific integrations from standard Kubernetes behavior and supports auditability of custom changes. When updating VKS clusters, the platform performs pre-checks and enforces compliance with compatibility requirements, reducing the risk of interoperability failures from format or version mismatches.

Identity federation uses standard protocols: VCF SSO supports SAML 2.0 and OpenID Connect (OIDC) for integration with external identity providers such as Okta, Microsoft Entra ID, Ping Identity, and Microsoft ADFS. VCF Automation also supports importing provider users from SAML, OIDC, or VCF SSO identity providers.

VCF Automation provides a structured blueprint format (formatVersion 2) with defined inputs and resources sections for specifying virtual machine and infrastructure component configurations. Cloud templates can be versioned and released to a catalog, with a dedicated API endpoint for creating new versions that captures change log entries, descriptions, release status, and version numbers. This versioning capability provides a mechanism for documenting changes to deployment templates over time. VCF Automation also supports importing AWS CloudFormation templates, allowing cloud administrators to create catalog items from both VCF-native templates and CloudFormation templates for deployment across cloud vendor regions or local datastores.

VKS Cluster Management Security Policies support export of custom policy template definitions in YAML format for offline editing, version control, and re-import across the VKS cluster estate, accessible through either the VCF Automation API or portal interface. This capability applies the same change-documentation approach used for cloud templates to Kubernetes security policy configurations, enabling organizations to track and review customizations to cluster security posture as versioned, auditable artifacts. VKS cluster management also provisions add-ons using standardized defaults when the Latest compatible version option is selected, applying consistent configuration baselines across the cluster estate; platform teams can standardize on a curated collection of add-ons to maintain operationally consistent VKS clusters across the environment.

VCF Operations tracks configuration drift across vCenter instances and vSphere clusters, comparing current settings against desired-state templates. Configuration drift templates can be integrated with source control repositories for versioning, auditing, and change control with reviews. Drift reports show historical changes by date, resource, and template, and administrators can trace configuration changes back to specific users and timestamps.

While VCF provides these standardized format capabilities, container interoperability, identity federation, and change-tracking mechanisms, the control also requires organizational governance processes to mandate the use of industry-recognized formats, review custom changes, and enforce interoperability requirements with cloud providers. These policy and procedural requirements fall outside the scope of the platform itself.

VMware vDefend (Contributes)

VMware vDefend contributes to business continuity by providing security-layer continuity mechanisms that remain active during disaster recovery events and workload migrations. NSX Federation Security maintains security posture across data centers during disaster recovery and workload migration, preserving distributed firewall and gateway firewall policies so that enforcement continues through failover.

The Security Services Platform, which provides the underlying infrastructure for vDefend's threat detection and inspection services, supports vSphere High Availability configurations. When a VMware ESX host becomes unhealthy, vSphere HA migrates the affected Security Services Platform virtual machines to healthy hosts in the cluster, restoring threat detection capacity without manual intervention. VMware recommends enabling vSphere HA for host-level failure recovery and configuring a storage policy with RAID for datastore-level failure tolerance in Security Services Platform deployments.

Security Services Platform can also be deployed on a vSAN stretched cluster configuration, which replicates storage across two physically distinct sites connected by a high-bandwidth or low-latency inter-site link. This configuration requires a

minimum of two sites with their own hosts and a witness host to handle potential conflicts arising from network partitions, supporting site-level failure scenarios.

When vSphere HA restarts workload VMs on a new recovery host after a failover, the vDefend distributed firewall automatically reinspects all traffic from those VMs on the receiving host, because the Flow Table contents cannot be moved between hosts during host recovery. Security policy enforcement resumes on the new host without manual reconfiguration.

The broader BCMS framework, including Continuity of Operations Plan and BC/DR playbooks, impact analyses, and organizational governance processes, is outside vDefend's scope and must be established through dedicated business continuity management programs.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides the technical execution layer for contingency planning in virtualized infrastructure. Recovery plans define the sequence, priority, and configuration for failover operations, acting as executable BC/DR playbooks. Protection groups organize virtual machines into logical units aligned with application tiers or business functions, establishing the scope of each contingency plan.

Recovery plans support non-disruptive testing before use in planned migration or disaster recovery, allowing organizations to validate their contingency procedures without affecting production workloads. Recovery plan configuration differs between emergency scenarios at the protected site and planned migration of services from one site to another, with PNR supporting both modes. A forced recovery function is available during disaster recovery plan execution for scenarios requiring immediate failover.

vSphere Replication provides host-based replication with configurable Recovery Point Objectives (RPO), allowing organizations to set per-workload data loss tolerances. vSphere Replication supports up to 200 point-in-time recovery instances per virtual machine using the Latest Instances option, enabling recovery to specific earlier states. This capability is relevant to contingency scenarios involving data corruption or ransomware events. When the source site is accessible at the time of recovery, vSphere Replication can synchronize recent changes from the source site to the target site before completing the operation, reducing data loss.

In 9.1, PNR extends to multi-tenant disaster recovery through integration with VCF Automation. The product implements role-based access control with Organization Administrator, Project Administrator, and Project Advanced User roles, each with defined authorization levels for disaster recovery operations. Role assignments must be configured on both the primary and secondary PNR sites to maintain consistent access control across the DR environment. A multi-region deployment topology places separate management domains in each region to contain disaster impact to one region, while enabling service recovery in an alternate management domain when VCF Automation and VCF Operations are affected. The VCF Operations Orchestrator Plug-In for PNR provides Configuration, Inventory Mappings, Protection Groups, and Storage predefined workflow categories for workflow-driven recovery operations.

Cyber recovery deployments support a standard VCF topology where the workload domain is isolated from the management domain, providing separation between management and user workloads in clean-room recovery scenarios.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) supports the use of secure protocols across its communication channels, contributing to organizational requirements for secure data handling in cloud-hosted database services.

For client-to-database connectivity, DSM database clusters use TLS certificates for both client connections and monitoring traffic. Clients connecting to a DSM-managed database cluster must add the cluster's certificate authority (CA) to the operating system or browser trust store to establish a TLS-encrypted session. The DSM UI and API both provide mechanisms to retrieve the CA for this purpose. DSM supports custom trusted certificate configuration; custom certificates must comply with FIPS requirements. DSM service components also use trusted certificates when connecting to external TLS-secured systems.

For connections to the Provider VM, DSM requires that the Provider VM's CA be added to the list of trusted CAs to establish a secure TLS connection. The CA must be stored in the trust store and used to validate the Provider VM certificate.

DSM applies secure protocol requirements across all supported database engines. PostgreSQL replication connections support SSL/TLS encryption with a configurable SSL mode (`sslmode` parameter) to secure data in transit between source and target databases. For ongoing client access, PostgreSQL databases support SSL-encrypted connections with downloadable CA certificates. MySQL replication operations include SSL trust validation: DSM verifies network access to the source database and validates SSL trust before proceeding. SQL Server clusters use TLS to encrypt communication between clients and the database cluster.

The Consumption Operator requires a TLS certificate for secure communication between the operator and the DSM provider. A Cloud Administrator must obtain and save this certificate when enabling self-service consumption of data services.

For outbound management communications, DSM supports Auto TLS configuration for SMTP server connections, available under Managing Log Forwarding Settings in the administration interface.

Access to data management operations is controlled through DSM's role-based access control model. Administrators have full management and monitoring capabilities across all data services, while users are scoped to data services they are authorized to access. This scoping restricts who can initiate import, export, or configuration operations within the platform.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides application-layer resilience mechanisms at both the control plane and the data plane that organizations can incorporate into business continuity plans for application delivery services.

At the control plane, the Avi Controller is deployed as a cluster with continuous configuration replication across all nodes, providing management-plane redundancy so that a single Controller node failure does not interrupt application delivery operations. Avi supports deployment of the Controller cluster across two availability zones, with vSphere HA recommended to support recovery of any node that fails in the secondary zone. For single-node deployments, enabling vSphere HA on the Controller VM is recommended for node recovery following a host failure. The Avi Controller provides configuration backup and restore capability that supports disaster recovery and configuration migration across Controller instances; organizations should capture and retain Controller backups as part of their continuity documentation.

Availability Zone awareness in Avi allows multiple availability zones to be configured, each defining a subset of vCenter instances, hosts, and datastores that share a fault domain. Configuring availability zones during Service Engine Group setup distributes data-plane capacity across isolated failure domains, supporting continuity designs where application delivery capacity remains available if one fault domain becomes unavailable.

At the data plane, Avi's Elastic High Availability mode provides fast recovery for individual virtual services following Service Engine failure. In vSphere DRS-enabled clusters, virtual service recovery occurs before vSphere HA brings up a replacement Service Engine when virtual service slots are available in other Service Engines, giving the application delivery layer its own recovery path distinct from the infrastructure restart cycle. If vSphere HA does not recover the failed Service Engine within a configured timeout period, the Avi Controller resumes recovery independently. Service Engine VMs should be deployed on vSphere clusters configured with both HA and DRS in fully automated mode; when Avi is deployed with VMware Kubernetes Service (VKS) on VCF, DRS and HA are required on every vSphere cluster used with Avi.

For Kubernetes environments, the Avi Multi-Cluster Kubernetes Operator (AMKO) provides federation architecture that supports disaster recovery across Kubernetes clusters. AMKO designates a leader cluster and multiple member clusters, coordinating application delivery failover as part of a multi-cluster DR strategy.

Avi ControlScripts, used in conjunction with the alert framework, can perform automated failover actions such as adding or removing routes based on VIP attachment and detachment events, allowing organizations to define automated remediation workflows that respond to system events without manual administrator intervention.

In 9.1 deployments using VCF Automation, Avi's Central Licensing maintains a global capacity pool that removes the requirement for duplicate licenses in DR scenarios, supporting license continuity across failover sites.

Organizations should document these mechanisms, including the Elastic HA configuration, Controller cluster topology and backup schedule, availability zone placement, and AMKO federation design, in their Business Continuity and Disaster Recovery playbooks. The technical recovery capabilities are built into the product; the planning artifacts, testing procedures, and escalation workflows are organizational responsibilities that extend beyond what Avi provides.

Control SOV-5

Supply Chain Sovereignty: Supply chain sovereignty evaluates the geographic origin, transparency, and resilience of the technology supply chain, focusing on the extent to which critical components and processes remain under EU control or exposed to non-EU dependencies. Contributing factors: Geographic source of key physical parts, manufacturing location - countries where hardware is manufactured or assembled; Jurisdiction and provenance of embedded code controlling hardware, firmware; Origin of Software: where and by whom software is architected and programmed, location and jurisdiction governing software packaging, distribution, and updates; Degree of reliance on non-EU vendors, facilities, or proprietary technologies; Visibility into the entire supplier and sub-supplier chain, including audit rights.

SCF Controls: AST-03.2, AST-13, MON-16.2, RSK-09, RSK-09.1, TDA-04.2, TPM-01, TPM-03

Aggregate Coverage: Not Applicable

Not applicable for this control: VCF, VMware vDefend, VCF Protection and Recovery, VMware Data Services Manager, VMware Avi Load Balancer. This control addresses an end user activity or process, a design decision, or activity outside the scope of these products.

Control SOV-6

Technology Sovereignty: Technology sovereignty evaluates the degree of openness, transparency, and independence in the underlying technological stack, ensuring EU actors can interoperate, audit, and evolve solutions without lock-in to foreign proprietary systems. Contributing factors: Ability to integrate with other technologies through well-documented and non-proprietary APIs or protocols, extent to which the solution adheres to publicly governed and widely adopted standards, reducing dependency on single vendors; Whether software is accessible under open licenses, with rights to audit, modify, and redistribute, ensuring transparency and adaptability; Visibility into the design and functioning of the service, including architectural documentation, data flows, and dependencies; Degree of EU independence in high-performance computing capabilities, including processors, accelerators, and software ecosystems.

SCF Controls: AAT-20.2, CFG-04.1, CLD-07, CLD-08, TDA-04.2

Aggregate Coverage: Contributes

VCF (Contributes)

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VMware vDefend (Contributes)

VMware vDefend contributes to secure protocol enforcement for the import, export, and management of data through certificate management, protocol governance, and communications security within the Security Services Platform.

The Security Services Platform uses certificates and secure communications protocols to protect transmissions between vDefend product components. When establishing external communications, the platform performs TLS handshake negotiation and actively rejects configurations that use deprecated algorithms, insufficient key lengths, or older protocol versions, applying this enforcement through FIPS-validated cryptographic modules.

The Security Services Platform's Security Segmentation capability identifies risky application protocols that transport data in clear text, and provides guidance for administrators on enforcing minimum secure protocol versions across the environment. The vDefend distributed firewall can then enforce network-level policy to restrict or block workload use of insecure protocols, limiting data movement to approved secure communication paths. Security policies applied to VPC environments are translated and enforced on transport nodes, providing consistent protocol controls across distributed infrastructure.

Certificate lifecycle management within the Security Services Platform supports importing certificates signed by customer-managed certificate authorities, enabling organizations to integrate vDefend into their existing PKI infrastructure. Certificate

consistency across Security Services Platform components, including backup and restore operations, is required to maintain uninterrupted secure communications for services such as Network Detection and Response sensors.

For integrations with external systems such as SIEM platforms, SSL verification is recommended to remain active for Network Detection and Response SIEM integration to reduce the risk of interception and modification of security telemetry in transit. Administrators can also configure whether Network Detection and Response flow data is shared to the cloud for advanced threat analysis, providing governance over what security data leaves the on-premises environment.

While vDefend secures its own component communications and provides network-level protocol governance for workload traffic, the broader scope of secure protocols across all cloud infrastructure layers, including storage encryption, management API security, and VM migration, is addressed by other VCF components including VMware vCenter and vSAN.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) uses industry-recognized formats and standards throughout its architecture, contributing to the interoperability requirements this control addresses.

For storage integration, PNR's Virtual Volumes support relies on VMware APIs for Storage Awareness (VASA). VASA providers, developed by third-party storage vendors, mediate communication between VMware vCenter and the underlying storage system. PNR requires that these VASA providers be registered in vCenter before a Virtual Volumes environment can be used with PNR operations. Storage characteristics reported by the Virtual Volumes provider appear in the VM Storage Policies interface, allowing administrators to create storage policies that reflect the capabilities of the underlying storage hardware. Through this VASA interface, Virtual Volumes supports the full range of PNR operations: replication, test recovery, test cleanup, planned migration, disaster recovery, and reprotect.

PNR exposes a REST API Gateway with documented endpoints for configuration and automation. The Protection and Recovery Server REST API provides endpoints to export configuration on demand and adjust export settings. A Configuration Import/Export Tool with schema support enables programmatic configuration management across sites, supporting consistent configuration of PNR infrastructure.

PNR integrates with VCF Operations through the VCF Operations orchestrator Plug-In for Protection and Recovery. This plug-in provides automated workflows to convert virtual machines to replicated state, manage IP customization rules, and administer PNR infrastructure through the VCF Operations orchestrator's standard workflow engine.

PNR deployments can span multiple cloud regions and include both on-premises and cloud-based site pairs, supporting recovery operations across hybrid environments that may involve multiple cloud providers or data centers.

The underlying VCF platform provides additional standardization through OVF template management, VMware Kubernetes Service API support, SAML 2.0 and OIDC identity federation, CNI-compatible container networking, and configuration drift tracking. These are VCF platform capabilities rather than PNR-native features.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) supports the use of secure protocols across its communication channels, contributing to organizational requirements for secure data handling in cloud-hosted database services.

For client-to-database connectivity, DSM database clusters use TLS certificates for both client connections and monitoring traffic. Clients connecting to a DSM-managed database cluster must add the cluster's certificate authority (CA) to the operating system or browser trust store to establish a TLS-encrypted session. The DSM UI and API both provide mechanisms to retrieve the CA for this purpose. DSM supports custom trusted certificate configuration; custom certificates must comply with FIPS requirements. DSM service components also use trusted certificates when connecting to external TLS-secured systems.

For connections to the Provider VM, DSM requires that the Provider VM's CA be added to the list of trusted CAs to establish a secure TLS connection. The CA must be stored in the trust store and used to validate the Provider VM certificate.

DSM applies secure protocol requirements across all supported database engines. PostgreSQL replication connections support SSL/TLS encryption with a configurable SSL mode (`sslmode` parameter) to secure data in transit between source and target databases. For ongoing client access, PostgreSQL databases support SSL-encrypted connections with downloadable CA certificates. MySQL replication operations include SSL trust validation: DSM verifies network access to the source database and validates SSL trust before proceeding. SQL Server clusters use TLS to encrypt communication between clients and the database cluster.

The Consumption Operator requires a TLS certificate for secure communication between the operator and the DSM provider. A Cloud Administrator must obtain and save this certificate when enabling self-service consumption of data services.

For outbound management communications, DSM supports Auto TLS configuration for SMTP server connections, available under Managing Log Forwarding Settings in the administration interface.

Access to data management operations is controlled through DSM's role-based access control model. Administrators have full management and monitoring capabilities across all data services, while users are scoped to data services they are authorized to access. This scoping restricts who can initiate import, export, or configuration operations within the platform.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer provides SSL/TLS termination and re-encryption capabilities that address secure protocol requirements for application-layer data in transit. Each virtual service that handles secure connections requires an SSL/TLS profile, which administrators configure to specify the cipher suites and protocol versions the service will accept. This makes protocol enforcement an explicit, per-service configuration that must be defined when deploying application services on Avi.

Avi operates as a proxy: client-facing connections terminate at the Service Engine, and the Service Engine re-encrypts traffic to backend application servers using a separate SSL/TLS configuration. When Service Engine-to-server re-encryption is enabled, the Service Engine can present a client certificate to backend servers, supporting mutual authentication on the downstream connection. This dual-termination model supports organizations in applying protocol requirements at both the client-facing and server-facing legs of an application connection, and in enforcing the requirement that all forwarded requests remain encrypted end-to-end.

In VCF-integrated deployments, the Avi cloud connector supports a `verify_certificates` option that uses the PKI profile with the System Trust Store to validate certificates when establishing connections to NSX Manager and vCenter. The Truststore PKI Profile flag must be set on the cloud configuration to enable certificate validation for cloud connectors. This extends secure protocol requirements to the control-plane connections between Avi and its VCF infrastructure dependencies, not only to application traffic.

For Kubernetes workloads managed through the Avi Kubernetes Operator (AKO), PKIProfile can be configured on ingress routes to enable certificate-based authentication and encryption for backend service communication. This allows the same SSL/TLS enforcement model to apply to cloud-native application traffic routed through Avi.

Control SOV-7

Security & Compliance Sovereignty: Security & Compliance sovereignty measures the extent to which security operations, compliance obligations, and resilience measures are controlled within the EU, ensuring independence from foreign jurisdictions and long-term operational assurance. Contributing factors: Attainment of EU and internationally recognized certifications (e.g., ISO, ENISA schemes); Adherence to GDPR, NIS2, DORA, and other EU frameworks; Security Operations Centres and response teams operating exclusively under EU jurisdiction, control over security monitoring/logging - customer or EU authority ability to oversee logs, alerts, and monitoring functions directly; Transparent, timely, and EU-compliant reporting of breaches or vulnerabilities, maintenance Autonomy - ability to develop, test, and apply security patches independently of non-EU vendors; Capacity for EU entities to perform independent security and compliance audits with full access.

SCF Controls: CPL-01, CPL-03, CPL-03.1, IRO-10, IRO-10.2, MON-01, MON-06, VPM-02, VPM-05

Aggregate Coverage: Contributes

VCF (Contributes)

VCF provides enterprise-wide monitoring through integrated tools that span infrastructure health, performance metrics, compliance assessment, log analysis, and alerting.

VCF Operations continuously monitors the operational state of all VCF components to detect conditions that may affect availability or compliance. It collects metrics across compute, storage, and networking resources, and correlates them over time to identify trends that may lead to potential failures. VCF Health, a component of VCF Operations, continuously monitors the operational state of VCF components to verify they are functioning as expected, giving platform operators a single view of operational status. Platform operators can configure alerts, perform log analysis, run infrastructure diagnostics, and review VCF health from a unified interface.

VCF Operations provides continuous compliance management by monitoring infrastructure against industry-standard benchmarks and custom policies. Administrators can create custom compliance benchmarks that monitor selected alert definitions and policies, tailoring the monitoring posture to organizational requirements. The platform assesses objects based on selected compliance policies and activates alert definitions associated with the current scorecard, providing ongoing visibility into compliance status. Compliance-related alerts are triggered on objects and displayed through a dedicated compliance alert subtype.

The alerting subsystem in VCF Operations supports the full lifecycle of alert management. Administrators can configure and manage alert definitions to monitor infrastructure events and conditions, including performance metrics, log events, and custom conditions configured through the Management Pack Builder. The Platform Health Alerts page provides a centralized view of all alerts related to overall system health, including alerts from data sources and infrastructure nodes. Health alerts are generated for conditions that affect the health of the environment and require immediate attention. Administrators can create policies to govern how alert definitions are evaluated, including which objects to monitor and what thresholds to apply, and can apply consistent practices to optimize alert behavior across monitored objects.

VCF Operations supports flexible grouping and scheduling to align monitoring with operational realities. Custom groups can be configured with automatic membership criteria to support monitoring strategies and achieve a high level of automation. Maintenance schedules can be assigned to objects undergoing planned downtime to keep monitoring data accurate and avoid false alerts when an object is offline for scheduled maintenance.

VCF Log Management provides centralized log collection and analysis. It ingests logs from VMware vCenter, VMware ESX hosts, NSX, vSAN, VCF Automation, VCF Operations, VCF Identity Broker, VCF Operations for Networks, and VCF Operations HCX. VCF Operations and VCF Log Management must be integrated to implement the centralized log collection architecture. The platform provides dashboards and analytics for troubleshooting and security investigation. Log-based alert

definitions allow administrators to trigger notifications when specific log patterns appear across monitored systems. Log forwarding to external SIEM or syslog targets is configurable, with support for TLS encryption to protect log data in transit. The vCenter syslog forwarding API supports TCP, UDP, TLS, and RELP protocols. NSX Manager syslog supports TLS and requires port 6514 for encrypted log transmission.

At the vSphere layer, the alarm infrastructure integrates with events and performance counters to provide real-time alerting. The `MetricAlarmExpression` data object allows administrators to set alarms on specific performance metrics with configurable monitoring intervals and aggregation functions for processing data points. The vSphere Client provides direct visibility into tasks and events for change tracking and troubleshooting. VCF ships with preconfigured vSphere alarms that are active by default and monitor specific components including the Identity Management Service, Message Bus Configuration Service, ESX Agent Manager, VMware Service Control Agent, vSphere HA host status, and host connection and power state. The vCenter Management Interface provides appliance-level health visibility through an Overall Health badge that reflects one of five states (Good, Warning, Alert, Critical, or Unknown) across all vCenter components, and notifies administrators of available security patches through Critical or Alert badge indicators when automatic patch checks are enabled.

When VCF Automation is integrated with VCF Operations, the Insights dashboard and Alerts tab provide real-time capacity awareness and alert information for VCF Automation objects, giving IT teams visibility into resource usage across both provisioned workloads and the underlying infrastructure. The VCF Automation Monitor tab surfaces deployment health information sourced from VCF Operations, and deployment cards display Last Request Status information that tracks the most recent operation on each deployment (Create, Update, or Delete) and its outcome (Successful, In Progress, or Failed). The Management Pack Builder allows administrators to set up event definitions for integrated systems, monitoring specific conditions or thresholds and configuring alerts for proactive management. Management Packs extend monitoring to network devices, including sensor monitoring for devices such as switches and routers. Adapters can be installed on VCF Operations nodes to extend monitoring to additional systems and data sources beyond the core VCF components.

The VMware Kubernetes Service (VKS) layer of VCF extends enterprise monitoring to container workloads and Kubernetes infrastructure. VKS clusters support `MachineHealthCheck` objects for both control plane nodes and worker node deployments; administrators manage these through the VCF CLI `vcf cluster machinehealthcheck` command, which supports configuration of unhealthy-condition definitions, maximum unhealthy thresholds (expressed as an absolute number or a percentage), and label selectors to target specific machines. The VCF CLI `vcf cluster machinehealthcheck node get` and `vcf cluster machinehealthcheck control-plane get` commands retrieve node and control plane health status on demand. VKS cluster health conditions, including the `Ready` and `ControlPlaneReady` status indicators, reflect the operational state of each cluster and are accessible via `kubectl` and the VCF CLI. The VCF Automation Provider Management UI includes a Services overview interface that exposes VKS cluster management functions and the health status of all cluster elements, giving platform operators a centralized view of the VKS tier. The Kubernetes API server within VKS clusters exposes diagnostic and health check endpoints (`/healthz`, `/livez`, `/readyz`, `/statusz`, and `/metrics`) that external monitoring systems can scrape to track API server availability and readiness; the `kube-apiserver` also emits the `kubernetes_healthchecks_total` counter and `kubernetes_healthcheck` gauge metric, which report structured health check status for components including `etcd` and `autoregister-completion`. Access to these monitoring endpoints is governed by the built-in Kubernetes `system:monitoring` RBAC group, which grants read access to liveness, readiness, and metrics endpoints while restricting broader cluster access. The Kubernetes Watch API allows monitoring tools to track changes to any API object as a live stream, supporting event-driven monitoring without continuous polling. The Node Problem Detector monitors node health and surfaces node-level problems as Kubernetes events and node conditions that external monitoring systems can consume via the event stream.

For workload-level monitoring within VKS clusters, Prometheus Operator is available as a standard package and defines custom resource definitions to deploy and manage Prometheus and Alertmanager `StatefulSets`, enabling high-availability alerting configurations. The Prometheus Alertmanager custom resource definition configures alert grouping, inhibition, and routing to external notification systems. The Istio service mesh, also available as a standard package on VKS clusters, provides service-level monitoring alongside its traffic management capabilities. The Dynatrace Operator can be deployed on VKS clusters to automate the deployment, scaling, and maintenance of the Dynatrace monitoring stack. Historical analysis, dashboarding, and alerting on cluster-level metrics require a dedicated monitoring solution configured to scrape the VKS

metrics endpoints and subscribe to the Kubernetes event stream. The vcf cluster support-bundler create command gathers diagnostic information from Supervisor and VKS clusters when troubleshooting is needed.

At the network layer, NSX provides monitored indicators including node uptime status, deployment status, management and controller connections, tunnel status, PNIC status, and uptime duration. NSX time series APIs allow viewing data such as VPN statistics, including metadata about the number of tunnels configured and how many are up or down at a given time.

VCF Operations also supports self-monitoring through the VCF Operations adapter, which collects internal metrics such as activity success counts. This allows administrators to verify that the monitoring platform itself is functioning correctly. The VCF Operations policy engine monitors conditions and generates events to react to changing states in the orchestrator server or connected technologies. The Adapter Instance monitoring interval is configurable, with intervals as granular as five minutes.

VMware Private AI Services (PAIS) includes a built-in observability stack that supplies monitoring data for AI infrastructure and AI workloads. PAIS Controller metrics are exposed at a Prometheus metrics endpoint on the pais-controller-manager service, allowing collection by monitoring systems and visualization in observability platforms such as Grafana. Observability is activated by updating the PAISConfiguration custom resource with optional sections for Prometheus metrics collection and LLM trace forwarding. The Prometheus runtime can be configured with a specified storage class policy and a configurable metrics retention period, giving administrators control over how long monitoring data is stored. A Prometheus server pod must be in a ready and running state for observability to function.

LLM trace data from PAIS can be forwarded to an OpenTelemetry Collector endpoint over http/protobuf or gRPC, providing distributed tracing for AI model inference activity. The observability.llmTraces configuration governs trace forwarding and works alongside the Prometheus metrics pipeline to produce two signals: metrics for health and performance, and traces for inference-level behavior. PAIS Controller metrics include the Model Endpoint Controller and PAIS Configuration Controller, giving monitoring platforms component-level visibility into PAIS operations.

In 9.1, PAIS adds direct integration with VCF Operations. The controller-pod-metrics-streaming setting activates streaming of PAIS controller pod metrics from the Supervisor cluster to VCF Operations, allowing administrators to view PAIS metrics alongside other VCF infrastructure metrics. A Telegraf deployment in the vmware-system-monitoring namespace uses a telegraf-user-config ConfigMap to manage user-configured monitoring endpoints for PAIS. The VCF Consumption CLI connects to the Supervisor endpoint and enables VI administrators to stream metrics from the PAIS controller pod. DevOps engineers can visualize health and performance metrics for PAIS components running in a VCF Automation namespace using Grafana, and MLOps engineers and application developers can monitor the health, quality, and behavior of AI applications and the models used in agents through the same observability platforms.

PAIS supplies the technical monitoring mechanisms this control requires. Enterprise-wide orchestration of monitoring controls, including alert threshold definition, escalation procedures, and integration with a centralized monitoring or log management platform, remain organizational responsibilities.

VMware vDefend (Contributes)

VMware vDefend provides event log generation and monitoring capabilities across its security components in 9.1, directly supporting anomaly detection and assessment within the network security domain.

The IDS/IPS subsystem generates detailed security event logs for event analysis. Remote syslog must be configured on ESX hosts to receive IDS/IPS events exported directly from those hosts; the `ids_events_to_syslog` setting controls whether IDS/IPS events are forwarded to external syslog consumers (the default is disabled). vDefend components write log data to `/var/log`, with syslog messages conforming to RFC 5424 on vDefend appliances and RFC 3164 on ESX hosts, enabling consistent ingestion by syslog-compatible platforms. The Security Explorer Computes view in Security Intelligence provides a full record of all IDS/IPS events for analysis over a 30-day timeframe. Security Intelligence also provides a View Full Event History capability that displays all events associated with a given signature ID along with their intrusion severity. The Threat Event Monitoring dashboard in Security Services Platform displays the trend in intrusion events over time, categorized by Intrusion Severity, giving security teams visibility into detection activity patterns.

Malware Prevention Service generates structured event logs with detailed metadata per file analysis event. Event log fields include the verdict (BENIGN, malicious, or other classification status), a numeric risk score, malware classification, malware family, error codes and messages, HTTP domain, HTTP method, user agent, file upload status, submission status, analysis completion status, and gateway identification. File event statistics appear on the Monitor & Plan > Overview > Threat Monitoring dashboard in Security Services Platform, with graphs covering a configurable time period (defaulting to the last hour).

Network Detection and Response (NDR) generates and sends event logs to a SIEM endpoint when a detection event occurs, enabling integration with external correlation and reporting platforms.

The Security Services Platform Monitor & Plan Overview dashboard provides a view of security posture that includes the latest security segmentation report, flow trends, a summary of detected suspicious traffic events, pending actions, and current settings requiring attention. Security Intelligence generates bar graphs showing the number of anomalies detected, categorized by severity. The Policy Recommendations section tracks Security Intelligence recommendations with status indicators such as Ready to Publish, allowing security teams to act on recommended policy changes.

vDefend Identity Firewall Event Log Sources can be integrated with VCF Log Management for centralized logging and monitoring. VCF Log Management supports Top-N queries using Interactive Analytics, enabling report generation from the aggregated log streams that vDefend supplies. VCF Log Management handles organization-level scheduled report generation, while vDefend provides the network-security-specific event data and dashboards that feed those reporting workflows.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides monitoring capabilities specific to the disaster recovery domain and integrates with the broader VCF monitoring ecosystem through dedicated Protection and Recovery dashboards in VCF Operations. VCF Operations collects configuration, component health, and usage information from protection and recovery servers and site pairs, presenting this through an overview dashboard that displays all site pairs, configurations, health scores, and protection and recoverability status for sites and virtual machines. PNR provides default health alerts for protection groups, recovery plans, cloud file systems, recovery SDDCs, and protected sites through VCF Operations, and the dashboards surface alerts informing administrators about configurations and deployments that require attention. Separate dashboard views cover vSAN Snapshots and Replication monitoring, and an Inventory view provides health status, metrics, alerts, and network topology for all monitored objects.

Within its own domain, PNR provides event logging and configurable alarms for replication and recovery operations. Administrators can create alarms that trigger on specified event conditions, with support for email notifications, SNMP traps, or script execution in response. PNR monitors connections between sites by sending periodic ping requests, logs connection check events, and generates alarms when broken connections are detected. The alarm system generates warning-level alerts for expired snapshot schedules, replication errors such as missing VMs, and connectivity issues with protected and target sites. PNR also monitors resource consumption on the Protection and Recovery Server host and raises alarms when configurable resource thresholds are reached.

PNR includes a health monitoring service that checks each health finding every five minutes and records a timestamp at each check. Health findings are tracked with status history that can be reviewed over configurable time periods, defaulting to a 24-hour window. The Protection and Recovery Appliance supports syslog forwarding to a remote syslog server for log analysis and centralized monitoring, allowing appliance-level events to flow into existing log management infrastructure.

The Clean Room Orchestrator, part of PNR's Cyber Recovery capabilities, supports monitoring of clean room events, tasks, and alarms for recovery operations. Clean Room Orchestrator can forward events to VCF Operations, Microsoft Sentinel, or Splunk Cloud for centralized monitoring, supporting organizations that require DR event data to flow into enterprise SIEM and log management platforms as part of a broader monitoring strategy. The Clean Room Orchestrator integrates with endpoint detection and response tools, including CrowdStrike and Carbon Black sensor appliances, for threat detection within isolated recovery environments.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) provides event log generation and reporting capabilities that support detection and assessment of anomalous activities within the database-as-a-service layer.

DSM's primary audit interface is the System Audit Events view, accessible through the System Audit section in the Provider UI. This view is the only location in the DSM interface where administrators can track non-database-related operations and tasks. Each audit entry records a Component field identifying the DSM component that performed the operation, an Event Type field identifying the nature of the audited event, Event Details providing operation-specific information, Event Time in UTC, and a Status field indicating the outcome of the event. The Source field records either a user identifier (Email ID) for user-initiated events or "System" for events initiated by DSM without user intervention, enabling attribution of each audit record to a specific actor. Standard Kubernetes audit events are also captured through the Gateway Audit Logs capability.

Audit events can be exported in CSV or XLSX format using the ACTIONS dropdown menu in the System Audit Events view. Exports include all events visible to the requesting user regardless of page size, covering the Component, Event Type, Event Details, Event Time (UTC), Status, and Source fields. These exports can be brought into external reporting and analysis tools to support audit review and anomaly assessment.

For persistent retention, DSM archives gateway audit log files periodically to an S3 repository subpath when a default S3 provider log repository is configured. Audit log files are also stored locally on the Provider VM at `/var/log/tadm/provider/audit/` in files named with a gateway-audit-timestamp.log format, supporting direct access for investigation. Support bundles can be generated and downloaded to aggregate control plane and data plane logs from a database cluster for diagnosis. The Provider VM also includes a System Logs capability for managing system log collection.

DSM provides operational monitoring that can surface indicators of abnormal behavior. The Monitoring tab for database instances displays metrics including Active Connections, Data Disk Usage, Write Throughput, Read Throughput, Thread Resource Utilization, Slow Queries per Second, InnoDB Buffer Usage %, InnoDB Reads and Writes, and Command Reads and Writes, with the time range adjustable through the Time Range Last dropdown. The Alerts pane in the Monitoring tab surfaces triggered alerts. DSM raises metrics-based alerts when metrics are successfully collected from a database cluster, status-condition-based alerts for critical failure conditions including database engine degradation and backup failures, and real-time alerts for resource or connectivity issues to support root cause identification. Webhook-based alerting supports integration with external notification systems for threshold-triggered database monitoring events. DSM can also be monitored through VCF Operations for additional troubleshooting and status visibility.

Satisfying this control fully requires organizational processes to review exported audit data and monitoring output, establish baselines for normal activity, and investigate deviations. DSM does not include built-in anomaly detection, correlation, or security information and event management capabilities. Organizations typically forward DSM logs to a centralized log management platform for automated detection and reporting at scale.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) contributes to enterprise-wide monitoring through application-delivery-layer mechanisms built into the Avi Controller and Service Engines.

The Avi Controller monitors Service Engine metrics and sends alarms and notifications to administrators when resource utilization data exceeds defined thresholds. Administrators configure custom thresholds on Service Engine groups through the Resource Events tab in the Avi UI. The Alert feature processes system events and metrics against user-defined rules and generates alerts that trigger configured actions, including sending an email, a syslog message, or an SNMP trap. Alert notifications can be forwarded to remote syslog servers for integration with centralized log management and monitoring platforms.

ControlScripts extend this event-response capability: a ControlScript can alter Avi configuration or communicate to external systems, including through webhook integrations with external monitoring tools, in response to defined system events. This supports dynamic responses including resource scaling or automated notifications based on health scores from monitoring data.

The Avi Controller and Service Engines track health score changes and SSL certificate expiration events. Avi integrates with external monitoring tools such as Prometheus for metrics collection and analysis. User-Defined Metrics allow administrators to define and extract customized telemetry data for specific compliance or reporting needs through the API. Real-time metrics are available via API but deactivate by default after 30 minutes to conserve resources and must be reactivated when sustained real-time visibility is required.

At the data plane, Avi health monitors track the status of backend servers and Service Engines continuously. Service Engine health monitoring records detailed server operational status, including health monitor name, type, last transition timestamps, rise count, fall count, total checks, and total failed checks. Each health monitor can be assigned its own threshold and timeout period for server health validation. Monitors support HTTP, HTTPS, TCP, and UDP health-check protocols with configurable success and failure thresholds, send intervals, and receive timeouts.

In Kubernetes environments managed by the Avi Kubernetes Operator (AKO), the HealthMonitor custom resource definition (CRD) allows users to define custom health monitoring configurations for backend services with fine-grained control over health check parameters. The Avi Multi-Cluster Kubernetes Operator (AMKO) GlobalDeploymentPolicy supports health monitor references for consistent monitoring across multi-cluster deployments. Within VCF Automation, a HealthMonitor custom resource can create monitors for tracking backend pool member status using TCP, PING, and HTTP check types.

Control SOV-8

Environmental Sustainability: Environmental sustainability assesses autonomy and resilience of cloud services over the long term in relation to energy usage, dependency and raw material scarcity.

Contributing factors: Adoption of energy-efficient infrastructure (e.g., low PUE) and measurable improvement targets; Circular economy practices ensuring reuse, refurbishment, and responsible end-of-life treatment of hardware; Transparent measurement and disclosure of carbon emissions, water usage, and other sustainability indicators; Sourcing of renewable or low-carbon energy to power infrastructure and operations.

SCF Controls: AAT-17.2, CAP-04

Aggregate Coverage: Contributes

VCF (Contributes)

VMware Private AI Services (PAIS) provides PAIS-native mechanisms that help organizations assess the environmental footprint of AI workloads. Metrics collection for PAIS is activated by uncommenting the `spec.observability.prometheusRuntime` section of the `PAISConfiguration` custom resource, after which DevOps and MLOps engineers can visualize PAIS health and performance metrics for components running in a VCF Automation namespace through observability platforms such as Grafana. The same observability pipeline lets MLOps engineers and application developers monitor models used in agents, giving operators visibility into the operational characteristics of individual AI workloads across deep learning VMs, VKS clusters, and RAG applications. This telemetry is the raw operational data that organizations use to quantify the energy and resource footprint of AI activity.

PAIS also supports resource efficiency through deployment leases. A lease controls how long a PAIS deployment can exist, so that AI workloads provisioned from VCF Automation catalog items are reclaimed at the end of a defined period rather than continuing to consume GPU and compute resources indefinitely. Disconnected (air-gapped) and connected deployment modes both rely on the same observability and lease controls, so the same sustainability-relevant data and lifecycle controls apply regardless of connectivity model.

Broader sustainability and green-metric monitoring at the cluster and host level is a VCF platform capability and is covered under VCF Coverage. The organizational activity of setting sustainability targets, analyzing collected metrics, and documenting environmental impact is a process obligation outside PAIS itself, which is why the rating is Contributes rather than Meets.

VMware vDefend (Contributes)

VMware vDefend provides automated monitoring of its own security infrastructure components through several complementary mechanisms. The Security Services Platform (SSP) Metrics service gathers performance data visualized across the Platform dashboard. The Metrics API supports time-series queries with filtering by metric names, object IDs, resource type, and granularity, and collects metrics including CPU load, network throughput, edge node datapath CPU utilization (`edge_dp_cpu.avg_percentage_busy_cycles`), and system resource counters. Flow collection provides additional visibility into system performance and resource utilization. The metrics service supplies data for Top N analysis, lifetime counters, and historical trend statistics to support firewall operational troubleshooting. For IDS/IPS operational health, the `get ids engine global stats` command provides a snapshot of the IDS engine's current performance and configuration state on a specific host.

SSP monitors deployment status and generates platform-related alarms highlighting errors requiring attention. When allocated resources exceed predetermined threshold values, SSP generates an alarm to alert administrators. SSP instance health status appears as Yellow or Red when subcomponent issues require investigation. On the firewall side, an alarm is triggered when dropped threat events are detected and remains active until no dropped events are detected, keeping administrators informed of policy enforcement gaps. Malware Prevention service deployment issues surface as alarms in the NSX Manager UI. The License Hub dashboard highlights platform-related alarms displaying system alerts and critical errors requiring attention.

SSP monitoring tracks the status of the flow clustering job, which runs every hour, reporting status values including COMPLETED, FAILED, SUBMISSION_FAILED, FAILING, INVALIDATING, NOT_AVAILABLE, PENDING_RERUN, RUNNING, and SUBMITTED. Security Intelligence suspicious traffic detectors report execution status after each run, with values of NOT_STARTED, SUCCESS, NOT_ENOUGH_BASELINE, or FAILURE.

The SSP Inventory monitoring capability displays the infrastructure hierarchy including regions, zones, environments, and virtual machines with operational status indicators that surface success, timeout, and out-of-sync conditions. Platform Inventory provides Power State information identifying whether a VM is running, stopped, suspended, or unknown. SSP displays health status for all onboarded Bare Metal servers, including BMS ID, BMS Name, Installer Version, Overall Status, and component-level details such as Control Connections and Distributed Firewall enablement status. The CLI provides a command to display the health of all services and agents running on a Bare Metal Server.

Network Detection and Response (NDR) can be integrated with SIEMs and generates event logs when detection events occur, campaigns are created, or campaigns are updated. SSP SIEM integration sends campaign and event logs to connected SIEM solutions including VCF Log Management and Splunk as events occur.

vDefend's monitoring scope is limited to its own security platform components. Centralized monitoring and alerting for the broader organizational Technology Assets, Applications, and Services inventory required by CAP-04 requires separate monitoring tooling outside of vDefend.

VCF Protection and Recovery (Contributes)

VCF Protection and Recovery (PNR) provides automated monitoring and alerting capabilities scoped to disaster recovery and replication health, contributing to centralized visibility into the operating state of critical recovery infrastructure.

PNR monitors connections between protected and recovery sites by sending periodic ping requests to detect broken connections and logging connection check events. When connectivity issues are encountered, or when protection groups and recovery plans enter an error state, alarms are generated automatically. PNR also monitors resource consumption on the Protection and Recovery Server host and raises an alarm if a resource threshold is reached; both threshold values and alarm behavior are configurable. Alarms can be configured to send email notifications, SNMP traps, or run scripts on the vCenter host in response to events. For email notifications to function, Mail settings must be configured in the vCenter Settings menu and an administrator email address must be specified during appliance registration.

vSphere Replication generates informational, warning, and error events that enable operators to monitor replication system health and resolve potential problems. Administrators can define and edit alarms that trigger when specific replication events occur, such as when a virtual machine is configured for replication or enters an error state. PNR dashboards display a historical chart of bandwidth consumed by replication tasks relative to available capacity, as well as VM utilization trends over time to identify virtual machines at risk of performance degradation.

PNR provides default health alerts for protection groups, recovery plans, cloud file systems, recovery SDDCs, and protected sites through VCF Operations. The VCF Operations Management Pack for PNR tracks server pairing status, server connectivity, recovery plan execution states, recovery plan errors, and protection group status, integrating DR-specific health metrics into a broader centralized monitoring environment. In environments with more than one vCenter, PNR displays events from all Protection and Recovery servers registered as extensions. PNR also tracks recovery history metrics including error count, execution time, result state, warning count, and powered-on and powered-off VM counts, providing a structured view of recovery plan performance over time. The Clean Room Orchestrator includes a monitoring interface that displays the operational status of configurations related to recoverability and supportability.

VMware Data Services Manager (Contributes)

VMware Data Services Manager (DSM) includes built-in performance monitoring and alerting capabilities for the database instances it manages and the infrastructure those instances depend on. Administrators and users can access monitoring data through the DSM Monitoring tab, accessible from both the vSphere Client plug-in and the standalone DSM UI. The Monitoring page displays real-time resource and performance metrics for each database VM, including Active Connections, Data Disk Usage, Write Throughput, Read Throughput, and CPU utilization. For MySQL instances, engine metrics are available including

Slow Queries per Second, InnoDB Buffer Usage, InnoDB Reads and Writes, and Command Reads and Writes. A dedicated Monitoring tab at the individual database instance level provides per-database visibility into operating state and health status.

DSM generates alerts at three severity levels: OK, WARNING, and CRITICAL. Administrators and users can configure the threshold values that trigger each severity for the key health parameters: CPU Health, Data Disk Health, System Disk Health, and Max Connections. Separate warning and critical thresholds are configurable per database instance through the Manage Alerts for Databases interface in the DSM UI, which allows teams to tune alerting sensitivity to operational requirements. DSM generates named alert types such as CPU_HEALTH_ALERT when CPU utilization exceeds a configured threshold.

In addition to database-level monitoring, DSM surfaces infrastructure-level conditions through a Global Alerts page visible in both the plug-in and standalone interfaces. Infrastructure alerts cover conditions such as VCENTER_NOT_REACHABLE (when the Provider appliance cannot reach VMware vCenter via the configured FQDN), VCENTER_CONNECTIVITY (when vCenter connectivity fails), INFRASTRUCTURE_POLICY_VC_ALARMS (when active vCenter resource alarms are present), orphaned VM detection, vCenterBinding API status, vCenter service account creation failures, and license-related conditions. These infrastructure-level alerts are paired with continuous, granular performance monitoring so that administrators can correlate resource conditions with database service impact.

DSM supports configurable notification routing through Webhook settings, configured under Managing Settings in the DSM UI. Webhooks transmit alert notifications for data service health events and can be configured to include alert level information indicating whether an alert is CRITICAL or WARNING, enabling integration with existing operational tooling and notification systems.

DSM introduces monitoring integration with VCF Operations. VCF Operations automatically discovers and monitors PostgreSQL and MySQL clusters in the Ready state, displaying performance data across three levels: cluster-level for overall health, node-level for infrastructure resource use, and database-level for database-specific KPIs such as Active Connections, Commits, Rollbacks, and Deadlocks. PostgreSQL-specific engine metrics include Commits and Rollbacks, Deadlocks and Conflicts, and Tuples Updated. MySQL-specific engine metrics include Queries and Questions, Bytes Received and Sent, and Slow Queries. Active alerts specific to individual nodes or the overall cluster are visible within VCF Operations. PostgreSQL databases support the pg_stat_statements extension to track execution statistics across all SQL statements, enabling query-level performance analysis.

For SQL Server databases managed by DSM, the monitoring framework supports metrics-based and status-condition-based alerts to identify critical failure conditions including database engine degradation, non-operational status, and automated or transaction log backup failure.

DSM provides automated monitoring and alerting for the database platform tier it manages directly. It does not function as an enterprise-wide centralized monitoring platform aggregating data from all technology layers, so organizations typically integrate DSM alerting and VCF Operations visibility with an enterprise monitoring solution and define policies for alert triage, escalation, and response.

VMware Avi Load Balancer (Contributes)

VMware Avi Load Balancer (Avi) includes a built-in Analytics subsystem that collects and presents performance metrics for virtual services, Service Engines, and backend pool servers. The Avi Controller Metrics service gathers this performance data and visualizes it across the Platform dashboard. The Analytics tab for virtual services displays metrics including end-to-end timing, traffic patterns, logs, and events. Service Engine analytics provides historical charts of metric tiles for the current virtual service, pool, or Service Engine, with configurable time periods ranging from the past 30 minutes to the past year. Performance data is accessible through the Controller UI and via the REST API, which returns aggregated metrics at configurable intervals, enabling external monitoring systems to ingest Avi operational data programmatically. The Avi for VCF guide confirms that Virtual Service Analytics captures and reports logs and events for application performance monitoring in VCF deployments. Within API gateway use cases, the monitoring capability tracks complete API transaction logs including response time and end-user experience data.

Avi supports configurable metric-based and event-based alerting. The Alert feature processes system events and metrics against user-defined rules and generates alerts that initiate configured actions. Alert actions deliver notifications through five

channels: the local Controller UI, email, syslog, SNMP traps, and ControlScript for custom automated responses. SNMP notifications support both SNMPv2c and SNMPv3 protocols, and can be configured with TLS encryption and strict certificate verification for secure transmission to external trap receivers. Avi supports multiple SNMP servers for alert notifications. These notification destinations are configured in the Operations > Notifications page of the Controller. Custom metric-based alerts can be defined against specific thresholds. Status-change events, such as Virtual Service Down events, generate SNMP trap notifications automatically. Autoscale alert actions can also generate informational notifications via email, syslog, or SNMP traps when Service Engine scaling occurs in response to load changes. The Avi Load Balancer Metrics Scripted Input pulls performance metrics using the REST API, supporting integration with Splunk for centralized collection of performance data.

Avi Service Engine monitors and alerts on resource utilization thresholds including CPU usage, memory usage, disk usage, and packet buffer usage. Detailed per-process CPU metrics are available, covering process name, CPU usage percentage, and memory usage percentage for Service Engine processes. Authorized users can also connect directly to a Service Engine and run standard CPU utilization monitoring tools to observe performance metrics in real time.

Avi health monitors continuously check the operating state of backend pool servers by issuing health-check requests to each pool member. Health monitors can detect when a server has entered maintenance mode by evaluating specific response codes or response data returned by the server, allowing Avi to route traffic away from pool members in maintenance. HTTP, TCP, and UDP health monitors all support maintenance mode detection via configurable response strings or response codes using the `maintenance_response` and `maintenance_code` parameters.

The Avi Controller exposes its own operational metrics through a REST API endpoint, including CPU usage statistics and process-level performance data. This enables an external monitoring platform to collect and centralize health and performance data from the Avi management plane alongside data from other infrastructure components.

